

Mini-project 1

1. Consider the backward heat equation (with homogeneous Neumann conditions), i.e. the inverse problem of recovering the initial temperature x from the temperature y at time $T > 0$. The forward operator is

$$Kx(t) = \int_0^\pi x(s)k(t,s)ds \quad k(t,s) = \frac{1}{\pi} + \frac{2}{\pi} \sum_{m=1}^{\infty} e^{-Tm^2} \cos(mt) \cos(ms).$$

- (a) K is compact in $L^2(0, \pi)$. Show that K has the singular system $\{x_j, y_j, \mu_j\}_{j=0}^{\infty}$ with $x_0 = y_0 = 1/\sqrt{\pi}, \mu_0 = 1$, and

$$x_j(s) = \sqrt{\frac{2}{\pi}} \cos(js), \quad y_j(t) = \sqrt{\frac{2}{\pi}} \cos(jt), \quad \mu_j = e^{-Tj^2} \quad j = 1, 2, \dots$$

- (b) Let $T = 0.05$ and $n = 256$. On the uniform midpoint grid $s_i = (i - \frac{1}{2})h, i = 1, \dots, n$, with $h = \pi/n$, the operator K can be applied to the vector $\mathbf{x} = (x(s_i))_{i=1}^n$ using the FFT-based discrete cosine transform in SciPy:

```
import numpy as np
from scipy.fft import dct, idct

def heat_forward(x, T):
    """Apply K on the midpoint grid s_i = (i-1/2)*pi/n."""
    j = np.arange(len(x))
    c = dct(x, type=2, norm='ortho') # cosine coefficients
    return idct(np.exp(-T * j**2) * c, type=2, norm='ortho')
```

Obtain the matrix $A \in \mathbb{R}^{n \times n}$ approximating K by applying `heat_forward` to the unit vectors, and compute the SVD of A . Compare in a plot the singular values of A with those of K , and explain the differences.

- (c) Fix the function

$$x(s) = s^2(\pi - s)^2.$$

Plot (using `heat_forward`) the functions x and $y = Kx$. Perturb y by adding Gaussian noise d of relative size $\|d\|_{L^2(0,\pi)} / \|y\|_{L^2(0,\pi)} = 1\%$ to obtain $y^\delta = Kx + d$, and make a Picard plot of the singular values μ_j , the expansion coefficients $|(y^\delta, y_j)|$ and the reconstruction coefficients $|(y^\delta, y_j)|/\mu_j$. Use the Picard plot to decide for a truncation in a TSVD reconstruction; compute the reconstruction and compare the result to the true x .

- (d) Consider the problem of computing the average $\bar{x}$ of x , i.e., $\bar{x} = \frac{1}{\pi} \int_0^\pi x(s)ds$ from the data y^δ . Is that an inverse problem, and is it ill-posed?

2. Consider the Abel problem defined in $L^2(0, 1)$ by

$$Kx(t) = \int_0^t \frac{1}{\sqrt{(t-s)}} x(s) ds.$$

With a numerical discretization based on a quadrature rule, choose uniform meshes in s and t with $n = 512$ points to set up the matrix $A \in \mathbb{R}^{n \times n}$, $n = 512$, approximating K . Fix the function

$$x(s) = s^2.$$

- (a) Sample the function $x(s)$ as a vector x , compute an approximation Ax of the exact $y(t) = Kx(t)$, and compare the two. Sample the function $y(t)$ to obtain y , and generate random noise d of relative size $\delta = 5\%$ to obtain $y^\delta = y + d$.
- (b) Compute TSVD solutions $x^{(k)}$ for $k \in [1, 100]$ and Tikhonov solutions x_α for $\alpha \in (10^{-4}, 10)$ to $Ax = y^\delta$. Plot the relative errors $\|x - x^{(k)}\|_2 / \|x\|_2$ and $\|x - x_\alpha\|_2 / \|x\|_2$ as functions of k and α , respectively, and in a separate figure plot the best TSVD and the best Tikhonov solution together with the true x . Compare the two methods, and explain why the solutions drop toward zero near $s = 1$.
3. Let K be a compact operator from Hilbert space X to Hilbert space Y with singular system $\{x_j, y_j, \mu_j\}$. Assume that K is injective.

- (a) Let $\tilde{R}_\alpha : Y \rightarrow X$ be defined in the singular system for K as

$$\tilde{R}_\alpha y = \sum_j \frac{\mu_j^2}{\mu_j^3 + \alpha} (y, y_j) x_j, \quad \alpha > 0.$$

Show that $\{\tilde{R}_\alpha\}_{\alpha>0}$ is a regularization scheme, i.e. that $\tilde{R}_\alpha$ is bounded for each $\alpha > 0$ and $\lim_{\alpha \rightarrow 0} \tilde{R}_\alpha Kx = x$ for all $x \in X$.

(Hint: use the expansion above to consider $\|\tilde{R}_\alpha Kx - x\|^2$ and make use of the Plancherel theorem. Be careful when interchanging the limit $\alpha \rightarrow 0$ and the summation.)

- (b) Now let

$$\hat{R}_\alpha y = \sum_j \frac{\mu_j}{\mu_j^3 + \alpha} (y, y_j) x_j, \quad \alpha > 0.$$

Is $\{\hat{R}_\alpha\}_{\alpha>0}$ a regularization scheme? Justify your answer.

Studying with AI tools in Mini-project 1 and beyond

AI tools are permitted and can be useful learning aids. Use them to support your learning and deepen your understanding of the course material.

Good practices

- Work on problems before consulting an AI tool.
- Use AI tools to explore concepts, generate alternative explanations, and test your understanding.
- Prompt for hints, ideas and feedback rather than complete solutions.
- Use AI tools for Python syntax, plotting and debugging, and make sure any retained code is understood.
- Compare and check generated output with course theory, numerical experiments and your independent reasoning; it may contain errors that look plausible.
- Generated output is often lengthy. Treat it as raw material: extract the essential ideas and present them concisely in your own words.
- Explore alternative approaches. Mathematical and computational problems can often be solved in different ways, so treat AI-generated output as one possible approach, not as an authoritative or unique answer.
- Discuss ideas with fellow students and instructors. Explaining concepts to others is one of the best ways to learn and prepare for the oral exam.
- Add a brief note in the report stating which AI tool(s) were used and for what purpose.

Rule of thumb

Make sure that you understand and can explain the reasoning and conclusions (now and at the exam) without access to AI tools. If not, the work is not yet complete.

Important notes:

- Deadline for the MP1 is Monday October 5, 2026.
- All mini-projects must be approved in order for you to qualify for the exam.
- Your background is different, and therefore you can choose to put emphasis differently in your solution.
- Demonstrate in your solution your understanding of theory and computations.
- Discuss the work with fellow students; this improves your learning and prepares for the exam
- You may even hand in as a group; every group member is accountable for the entire solution.
- The number of pages for the main report must not exceed 7. Code can be put in an appendix.
- Comment on all results: What do you see, and why is it reasonable (or unreasonable)?
- Make quality plots with proper names on axis, legends and captions. Zoom in on the relevant stuff.
- You might already now prepare the presentation of the MP to be used for the exam.